Pablo Villacorta Aylagas

Telecommunications Engineer
PhD Candidate

pablo.villacorta@uva.es

May 29, 2000

Valladolid, Spain

Resume

Experience

MRI Researcher - PhD Candidate
Image Processing Lab (LPI) - Universidad de Valladolid
2022 - Present
Working on MRI simulation frameworks, with a focus on modeling complex phenomena like blood flow and non-rigid motion. Also involved in optimizing simulation pipelines and improving algorithms performance and accuracy.

Education

Universidad de Valladolid, Spain
MSc in Telecommunications Engineering
2022 - 2023
Habilitating MSc focused on real-world projects. We designed an IoT datalogger with LoRa and sensor integration, and deployed a 4G-based system for real-time video processing and AI traffic sign detection.

Universidad de Valladolid, Spain
BSc in Telecommunications Engineering
2018 - 2022
Acquired a solid foundation in signal processing, programming, electronics, systems engineering, and telematics.

Languages

English - C1 Level
Certified with Cambridge Certificate in Advanced English (CAE)
2026
Score: 198
Certified with Cambridge First Certificate in English (FCE)
2020
Score: 182

About me

I am a Telecommunications Engineer interested in all its areas, from signal processing to programming and telematics, including disciplines such as electronics and instrumentation. I am also passionate about Magnetic Resonance Imaging (MRI), a field that brings together the best of each discipline and offers room for innovation to improve people's health.

Currently, I am working as a researcher in the Image Processing Lab (LPI) at Universidad de Valladolid (Spain), where I contribute to the **KomaMRI.jl** project while developing a **web-based tool for MRI pulse sequence editing and simulation**.

Publications

Journal Papers

2026

A multi-*b*-value test-retest diffusion MRI brain dataset for model validation and reproducibility assessment
Sumbitted to Scientific Data. (Preprint) DOI: 10.64898/2026.08.23.746449
T. Pieciak, I. Guadilla, D. Ciupek, R. Navarro-González, S. Merino-Caviedes, **P. Villacorta-Aylagas**, L. Magdaleno Humayor, M. Villa Aparicio, J. Rueda-Ramos, R. Santiesteban-Mendo, R. Moro-Boyero, A. Tristán-Vega

MRSeqStudio: MRI Sequence Design and Simulation *as a Service* in a Free and Open-Source Web Platform
J Med Syst, 50, 65 (2026). DOI: 10.1007/s10916-026-02394-1
P. Villacorta-Aylagas, M. Rodríguez-Cayetano, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, C. Alberola-López

Versatile and Highly Efficient MRI Simulation of Arbitrary Motion in KomaMRI
Magn Reson Med, 95, no. 3 (2026): 1791–1803. DOI: 10.1002/mrm.70145
P. Villacorta-Aylagas, C. Castillo-Passi, R. Kierulf, R.M. Menchón-Lara, J.R. Rodríguez-Galván, J. Sierra-Pallaers, P. Irarrazaval, C. Alberola-López

Sampling of non-Gaussian Ensemble Average Propagators for the simulation of diffusion magnetic resonance images
Magn Reson Med, 95, no. 2 (2026): 1255–1265. DOI: 10.1002/mrm.70080
J.R. Rodríguez-Galván, **P. Villacorta-Aylagas**, S. Merino-Caviedes, F. Simmross-Wattenberg, C. Castillo-Passi, P. Irarrazaval, A. Tristán-Vega, C. Alberola-López

Conference Contributions

(N: National, I: International, P: Poster, O: Oral)

2026

Enhancing KomaMRI with a Pulseq I/O Interface for Reproducible Sequence Workflows (P)
2026 ISMRM Iberian Chapter Meeting, Coimbra, Portugal (I)
P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López
[Abstract] [Poster]

MRSeqStudio: A Free and Open-Source Web Application for MR Sequence Design and Simulation as a Service (P)
2026 ISMRM & ISMRT Annual Meeting & Exhibition, Cape Town, South Africa (I)
P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

2025

A Free and Open-Source Web Application for Pulse Sequence Development and Simulation (P)
1st Best Poster Presentation
2025 ISMRM Iberian Chapter Meeting, Barcelona, Spain (I)
P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López
[Abstract] [Poster]

2024

Flexible Motion Models for GPU-accelerated MRI Simulations with Complex Motion and Flow using KomaMRI (O/P)
ESMRMB 2024, Barcelona, Spain (I)
P. Villacorta-Aylagas, C. Castillo-Passi, R.M. Menchón-Lara, J. Anatol-Hernández, P. Irarrazaval, J. Sierra-Pallares, C. Alberola-López
[Abstract] [Poster]

Proof of Concept for the Simulation of Diffusion Models in the Myocardium through KomaMRI (O)
CASEIB 2024, Sevilla, Spain (N)
J. R. Rodríguez-Galván, **P. Villacorta-Aylagas**, S. Merino-Caviedes, C. Castillo, J. Sánchez-González, D. Filgueiras-Rama, A. Tristán-Vega, P. Irarrazaval, C. Alberola-López

New Methods for Defining Dynamic Phantoms in KomaMRI (P)
2024 ISMRM Iberian Chapter Meeting, Porto, Portugal (I)
P. Villacorta-Aylagas, C. Castillo-Passi, R.M. Menchón-Lara, J. Anatol-Hernández, P. Irarrazaval, J. Sierra-Pallares, C. Alberola-López

Synthetic spin trajectory generation for advanced Flow phantom in MRI simulation (P)
2024 ISMRM Iberian Chapter Meeting, Porto, Portugal (I)
J. Anatol-Hernández, J. Sierra-Pallares, **P. Villacorta-Aylagas**, C. Castillo-Passi, R.M. Menchón-Lara, P. Irarrazaval, C. Alberola-López

A KomaMRI Phantom Extension for Integrated Dynamic Imaging (P)
2024 ISMRM & ISMRT Annual Meeting & Exhibition, Singapore (I)
P. Villacorta-Aylagas, C. Castillo-Passi, R. M. Menchón-Lara, P. Irarrazaval, J. B. Sierra-Pallares, C. Alberola-López

2023

A Web Version of KomaMRI. Sequence Editor and Remote Execution (P)
2023 ISMRM Iberian Chapter Meeting, Valladolid, Spain (I)
P. Villacorta-Aylagas, I. Fernández-Arias, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez Cayetano, C. Alberola-López

Parallel Device-Agnostic Simulation of Magnetic Resonance Imaging Sequences (P)
2023 ISMRM Iberian Chapter Meeting, Valladolid, Spain (I)
I. Fernández-Arias, **P. Villacorta-Aylagas**, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

2022

A Cross-Plattform Editor and Simulator of Magnetic Resonance Imaging Sequences. Design and Implementation (O)
CASEIB 2022, Valladolid, Spain (N)
P. Villacorta-Aylagas, I. Fernández-Arias, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez Cayetano, C. Alberola-López

Simulación paralela multidispositivo de secuencias de imagen por resonancia magnética (P)
CASEIB 2022, Valladolid, Spain (N)
I. Fernández-Arias, **P. Villacorta-Aylagas**, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

Projects

Competitively funded research projects

PID2025-169651OB-I00

Cálculo automático de mapas de grosor auricular e implicaciones en la caracterización del remodelado auricular

Granted (provisional resolution) · AEI / MICINN · Proyectos de Generación de Conocimiento 2025 · Investigación orientada

Research team member. PI: Carlos Alberola-López. Image Processing Lab (LPI), Universidad de Valladolid. Not started yet (pending definitive resolution / start date).

Contrato predoctoral UVa–Banco Santander

Simulación de resonancia para una transición digital en la formación y el entrenamiento de técnicos en imagen y radiólogos

2025 – 2029 · Convocatoria 2024 de contratos predoctorales UVa cofinanciada por Banco Santander

Predoctoral researcher. Universidad de Valladolid.

TED2021-130090B-I00 (TDSim)

Simulación de resonancia para una transición digital en la formación y el entrenamiento de técnicos en imagen y radiólogos

2022 – 2025 · AEI / MICINN · Proyectos estratégicos orientados a la transición ecológica y digital 2021 · NextGenerationEU

Hired as "Investigador en Formación". PI: Carlos Alberola-López. Image Processing Lab (LPI), Universidad de Valladolid.